

Motivation

- Leaky Integrate-and-Fire Neurons
 - work by integrating (accumulating) input currents over time to increase a "membrane potential" and "fire" (produce a spike) when this potential crosses a threshold, similar to how a leaky RC circuit charges. After firing, the potential resets. The "leaky" part refers to the fact that the membrane potential also gradually decays or "leaks" back to a resting value over time unless it is being driven by input
- The SCN idea
 - a population of LIF neurons only fire when it reduces the global representation error. This yields sparse, irregular, yet coordinated spikes
 - The paper shows that the same network can implement a linear dynamical system (add a "slow" recurrent term).
 - Kalman filter for optimal state estimation through noisy observations
 - LQR for optimal control to a target $z(t)$
 - The control u is a linear readout of the spikes
 - -> Spiking LQG

Setup

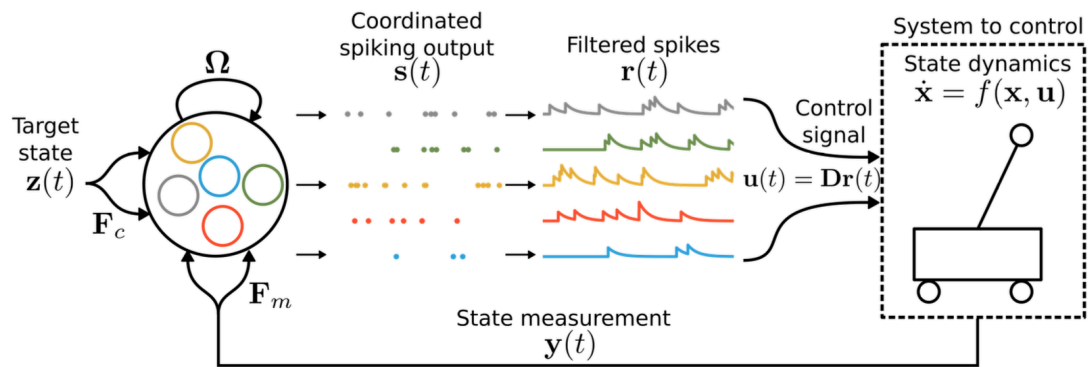


Fig. 1: Controlling dynamical systems with spiking neural networks. A recurrent spiking neural network (left) will emit spikes based on its inputs and connectivity, which are translated into a control signal to control some target system (right). F_c and F_m represent input weights, Ω represent recurrent weights, and D represent read-out weights. By first defining the optimal control solution for the system, and then directly translating the resulting control parameters into these connection weights, we can generate a robust and population-wide coordinated spiking network that accurately controls identified systems.

- x : state
- y : incomplete measurement of the system state
- z : incomplete measurement of the target state
- u : control signal
- s : spiking signal emitted from a recurrent SNN, used to control the system: filtered spikes

Emitted Spikes and Filtered Spikes

- s : spiking signal emitted from a recurrent SNN, used to control the state

$$s_i(t) = \sum_k \delta(t - t_i^{(k)}), \quad s(t) = [s_1(t), \dots, s_N(t)]^\top.$$

○

$$s_i[n] = \begin{cases} \frac{1}{\Delta t}, & \text{if neuron } i \text{ spikes in bin } n, \\ 0, & \text{otherwise,} \end{cases}$$

○ discrete:

○ $s_i(t)$ is the sum of the spikes emitted from the neurons in time t .

• Each $t_i^{(k)}$ is the k -th spike time of neuron i .

• Integrating s_i over any interval counts spikes: $\int_{t_0}^{t_1} s_i(t) dt = \#\{\text{spikes of } i \text{ in } [t_0, t_1]\}$.

• This is a **distribution** (generalized function), not an ordinary continuous signal. In their LIF network, a spike occurs when the voltage $v_i(t)$ hits its threshold T_i ; then s_i contributes a delta at that instant.

○

- r : filtered spikes

$$\dot{r}(t) = -\lambda r(t) + s(t) \quad \Rightarrow \quad r = (e^{-\lambda t} H(t)) * s,$$

○ so each spike adds a unit jump to r that then decays with time constant $1/\lambda$.

Closed-form control with spike ...

$$\dot{r}(t) = -\lambda r(t) + s(t) \quad \Longleftrightarrow \quad r(t) = \int_{-\infty}^t e^{-\lambda(t-\tau)} s(\tau) d\tau.$$

○

- Dirac impulse delta function

$$\delta(x) = \begin{cases} 0, & x \neq 0 \\ \infty, & x = 0 \end{cases} \quad \text{with} \quad \int_{-\infty}^{\infty} \delta(x) dx = 1.$$

○

○ it's not a real function, so approximated with limits: $\delta_a(x) = \frac{1}{|a|\sqrt{\pi}} e^{-(x/a)^2}$ $a \rightarrow 0$

Decoding matrix

- Assumption

○ spikes should only be emitted when this improves the coding error defined by the L2-norm (yielding a GREEDY SPIKING RULE): neuron i should spike iff

$$\|x - Dr\|_2^2 > \|x - Dr - D_i\|_2^2$$

- Calculating Threshold

$$\|e\|_2^2 > \|e - D_i\|_2^2$$

$$\|e\|_2^2 > \|e\|_2^2 - 2D_i^T e + \|D_i\|_2^2$$

- $0 > 0 - 2D_i^T e + \|D_i\|_2^2$
- $D_i^T e > \frac{1}{2}\|D_i\|_2^2$
- $v_i > \frac{1}{2}\|D_i\|_2^2$
- so set the threshold T_i as such

Spike Coding Theory

- If a K-dimensional state $x(t) \in \mathbb{R}^K$ is fully observable without noise, SCN theory defines how a recurrent SNN can optimally track this signal
 - Assumptions
 - the input signal estimate $\hat{x}$ can be linearly decoded from the spiking activity as $\hat{x} = Dr$, where D in $\mathbb{R}^{(N \times K)}$ are known decoding weights (in this paper randomly drawn from the normal distribution and normalized)
 - spikes should only be emitted when this improves the coding error defined by the L2-norm (yielding a GREEDY SPIKING RULE): neuron i should spike iff $\|x - Dr\|_2^2 > \|x - Dr - D_i\|_2^2$
- The main equations
 - $\hat{x} = Dr$
 - $\dot{r} = -\lambda r + s$
 - $\Omega_f = -D^T D$
 - $\dot{v} = -\lambda v + D^T(\dot{x} + \lambda x) + \Omega_f s$